

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE SEMESTER-II , SECOND INTERNAL EXAMINATION, JULY 2024
ENGINEERING PHYSICS (23-200-0102A)

ME A

Max. Marks: 30

Time: 2hrs

Bloom's Taxonomy levels (BL) L1- Remember, L2- understand, L3- apply, L4- Analyse, L5 Evaluate, L6 -Create

Course Outcomes

On successful completion of the course, the students will be able to :

CO1: Interpret modern devices and technologies based on lasers and optical fibres.

CO2 : Explain the basic principles of crystal physics.

CO3 : Summarise the characteristics and applications of superconducting materials, nano materials and smart materials.

CO4 : Illustrate the theory of semiconductors and magnetic materials.

CO5 : Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART – A

Attempt all questions

(5 x2 = 10)

Q.No I.		Marks	BL	CO	PO
1	Describe the two phases in shape memory alloys.	2	1	3	2
2	Explain the term surface occupancy.	2	2	3	2
3	Prove that a material in the superconducting state is a pure diamagnet.	2	3	3	2
4	Differentiate between intrinsic and extrinsic semiconductors.	2	1	4	2
5	What is the significance of Neel temperature in magnetism ?	2	2	4	2

PART – B

Answer any 2 questions from the options provided.

(2 x10 =20)

Q.No.		Marks	BL	CO	PO
II (a)	With the help of suitable diagrams explain the details for the classification of superconductors.	7	2	3	2
(b)	Calculate the critical current flowing through a long thin superconducting material of diameter 2 mm, given critical magnetic field for the sample is 8×10^3 amp/m.	3	3	3	2
	OR				
III (a)	Explain the working of a twisted nematic liquid crystal system with suitable diagrams.	7	2	3	2
(b)	Give two advantages and disadvantages each, for liquid crystals over LED.	3	1	3	2

Q. No		Marks	BL	CO	PO
IV (a)	In detail bring out the differences between diamagnetic and paramagnetic materials with examples and applications (2 each).	7	2	4	2
(b)	Define intensity of magnetisation and magnetic permeability .	3	1	4	2
	OR				
V (a)	Explain the different types of semiconductors with examples and applications (2 each).	7	2	4	2
(b)	Bring out the effect of temperature on the working of a semiconductor	3	2	4	2

BL1 - 31 %,

BL2 – 54 %

BL3 - 15 %